

Integrated Machine Learning Driven and Soil Clustering Framework for Crop Recommendation in Cauvery Delta Region

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Abstract—The fertile Cauvery Delta region of Tamil Nadu is regarded as important to India's agricultural financial system. In contrast to the traditional methods for increasing yield, the importance of both Photosynthetic Active Radiation (PAR) and the pH factor are considered in this research study for increasing the yield of crops in the Cauvery Delta region. PAR is identified as the fraction of light utilized by plants, algae, and cyanobacteria for photosynthesis to transform light energy into chemical energy. PAR values are not often considered in most research papers due to their unavailability. In this study, six months of PAR readings were gathered from a real-time weather station in the region. Using linear regression, the study examined how PAR is linked to temperature, and how pH relates to nutrient levels in the soil. A Random Forest Regressor was also used to explore how PAR interacts with temperature, illuminance, and total solar radiation, with strong results ($R^2 = 0.99$). The PAR–temperature-only model had a lower R^2 of 0.79, and pH prediction reached 0.80. A TP–NP mapping method was applied to check how local greenhouse data compares with global standards. To further support decision-making, hierarchical clustering was used to group villages and soil types. This enabled the development of a region-specific crop recommendation system, which aligns local soil clusters with optimal crops for each village. The overall approach links real-time environment prediction with practical advice for farmers. It shows how PAR and pH can be included to support better greenhouse decisions and sustainable crop production.

Keywords: *Photosynthetically Active Radiation (PAR), pH Prediction, Total Solar Radiation (TSR), Sustainable Agriculture, Hierarchical clustering.*

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I. INTRODUCTION

The Cauvery delta region located in Tamil Nadu is known as the granary of the state [1] because it has rich alluvial soil and abundant water resources for optimal plant growth. This region contributes 30 percent of the food in the state, but the numbers have been reduced to 20% due to changes in climate observed in recent years. This reduced agricultural productivity is the most important concern for smart agricultural practices. Greenhouse agriculture offers a controlled environment which includes factors like temperature, humidity, light can be manipulated to achieve maximum plant growth. This provides crop production even in times of vulnerable climatic conditions.

Among the many factors which contribute to plant growth, PAR and soil pH are two of the most crucial factors. PAR refers to the portion of sunlight that plants use for the process photosynthesis, which converts light energy to chemical energy to use for food production. In greenhouse agriculture, controlling PAR, regardless of weather conditions guarantees that plants receive adequate energy for photosynthesis [2]. This reduces farmer's dependency on weather conditions for crop yield. In addition to PAR, soil pH is equally contributing to plant growth. Soil pH affects nutrient availability and determines whether plants can absorb the nutrients. When pH is too low, it creates an alkaline environment; plants in alkaline environment cannot absorb certain nutrients. By carefully selecting pH within the optimal pH range of Cauvery delta region crops and regulating PAR along with it will maximize crop yield.

To understand better the relationship between PAR and pH values in greenhouse cultivation, Table 1 shows the optimal ranges of PAR and pH required for tomato [4].

Table 1. Optimal Ranges of PAR and pH for Tomato

pH Range / Growth Suitability	Optimal PAR ($\mu\text{mol}/\text{m}^2/\text{s}$) / About Crop
0-4	Unsuitable / Not applicable / Extreme acidity; toxic for plant growth.
4-5	Marginally suitable / 400-500 / Improved nutrient retention, water holding capacity, and reduced stress effects.
5-6	Suitable / 400-600 / Water uptake, root development, and soil aeration are enhanced for improving nutrient availability.
6-7	Ideal / 500-700 / Optimal pH for nutrient absorption; promoting strong growth.
7-8	Suitable / 500-600 / Minor alkalinity impacts, supporting healthy growth.
8-10	Marginally suitable / 400-500 / Nutrient uptake is limited under saline stress conditions.
10-14	Unsuitable / Not applicable / Highly alkaline and toxic for growth.

From the Table I, the optimal pH and PAR are studied. These optimal conditions foster plant health, nutrient absorption and yield [5].

In the crop-growing scenario of Cauvery delta, both linear regression and random forest regression can be employed for understanding the impact of environmental factors such as PAR and soil pH on the yield. These Machine-learning algorithms are assisting in forecasting and discovering the best environment suitable for plant growing within a greenhouse because of the necessity of accuracy. Given current climate change tendencies, it is critical to use data analysis and assessment approaches to improve the utilization of the obtainable resources like PAR and pH to support and boost crop yields [6]. Linear regression explains how PAR and soil pH affect crop yield, disclosing the strength and direction of their influence. Such insight makes it easier to predict yields under different conditions. Random Forest regression, on the other hand, might be able to capture more complex relationships among temperature, humidity, and light and provide improved accuracy. Furthermore, By using six months of live PAR data from weather stations and soil testing results from the Soil Testing Laboratory in Aduthurai, this research constructs a predictive model for greenhouse cultivation. In addition, hierarchical clustering is used to cluster villages based on soil and environmental attributes to facilitate a region-specific crop

recommendation system. The combination of these models would achieve better decision-making for greenhouse gardening by tuning PAR and pH for better yields, supporting sustainable farming in the Cauvery delta region.

II. LITERATURE REVIEW

The study uses modeling techniques such as Linear Regression and Random Forest Regression to examine the relationship between plant growth in the Cauvery delta's greenhouse environment and important environmental factors, including PAR, temperature, pH, and soil nutrient composition. The objectives are focused on manipulating these variables, more particularly the PAR and pH to enhance the germination of plants and enhance the practice of sustainable agriculture in the region. The research works presented in this section are the key to the achievement of these objectives as these seek to establish the effects of environmental parameters such as PAR, pH and soil characteristics on plant health and growth.[7] and [8] deal with the application of machine learning for modeling estimations of parameters, which depend on PAR and soil characteristics, including chlorophyll content and biomass.[9] and [10] present the study on the estimation of fraction of PAR absorbed by plants using spectral indices and remote sensing which give instantaneous updated information on efficiency of light absorption by the plants in your green houses. The pattern of how soil pH influences the growth of plants, discussed in [11] and [12].

These studies demonstrate how growth media pH affects the nutrients available. The research work presented in [13] examines how prismatic density affects the effectiveness of light capture, another aspect of successful PAR for photo phytic processes. [14] and [15] discuss how, based on data, remote sensing intentions and a machine learning algorithm, the growth of the crops can be predicted, and the soil fertility can be monitored. These studies stress on how growth conditions can be controlled through monitoring. In [16] and [13] authors used multispectral data and machine learning to model some plant growth descriptors such as the LAI.

Furthermore, hierarchical clustering techniques have been explored in recent research

to group geographic regions or soil types based on environmental and nutrient similarities. This clustering approach [17] supports region-specific crop recommendations, improving land use efficiency and decision-making in precision agriculture. With the use of these techniques, models will come up with better predictions on the state of plant health and growth and therefore contribute positively to determining the right conditions to foster sustainable agriculture. In brief, the above research papers present valuable information for machine learning and remote sensing techniques and environmental monitoring systems as being useful for achieving the goal of enhancing PAR and pH for improved plant productivity. With these methodologies, there is the ability to design better strategies of managing the conditions in the green house in the Cauvery delta region and enhance the spread of efficient methods in agriculture.

III. OBJECTIVE OF THE RESEARCH WORK

The primary goal of this research work is to optimize crop yield in the Cauvery Delta region by integrating PAR, pH levels, and other environmental parameters into predictive models. The research focuses on the development of region-specific insights for greenhouse crop management and enhances sustainable agriculture practices. The main contributions of the proposed work are the following:

- To analyze the relationships between PAR, temperature, soil nutrient concentrations, and pH levels using machine-learning models, focusing on the Cauvery delta region.
- To use Linear Regression and Random Forest Regressor to estimate crop PAR and pH values using real-time meteorological data and soil test findings.
- To use expected PAR and pH values and compare them with global standards to optimize greenhouse environmental conditions, increase crop yield and encouraging sustainable agriculture.
- To develop a crop recommendation system using Hierarchical Clustering, which groups villages and soil profiles based on environmental and nutrient similarities, thereby enabling the selection of the most suitable crops for each

cluster. This data-driven approach ensures efficient land use and aligns cultivation practices with localized soil and climate characteristics.

IV. PROPOSED METHODOLOGY

The overall architecture is divided into three phases namely:

- Integrated Soil, Crop, and Weather Data Collection Framework.
- Predictive Modeling with TP-NP Mapping Logic.
- Soil Clustering and Crop Recommendation.

Figure 1 depicts the overall architecture of the proposed system.

A. Stage 1: Integrated Soil, Crop, and Weather Data Collection Framework

To implement the proposed model architecture, the respective dataset required for collected from the Public Works Department (PWD). Initial weather data, specifically for PAR, was obtained from real time weather station installed at MHI sponsored TBI at SASTRA Deemed University. The weather data utilized spans the period from March to September 2024.

The weather dataset comprises 11 major attributes and spans data for 6 months. Each day includes 143 data points, resulting in a total dataset size of $[26,030 \times 17]$. Subsequently, the soil dataset and crop dataset were acquired from Soil testing laboratory, Aduthurai. The weather, soil and crop data set attributes are given in Table 2, 3 and 4 respectively.

Table 2. Weather Dataset

S.NO	WEATHER ATTRIBUTES
1	Time
2	CO2
3	Humidity
4	Illuminance
5	PAR
6	Temperature
7	Total Solar Radiation (TSR)
8	Pressure
9	Wind Speed
10	Rain Accumulation
11	PPM10

The weather parameters were chosen based on their direct influence on photosynthesis (PAR), light interception (Illuminance, TSR), and climate responsiveness, aiding accurate crop environment modeling. The soil fertility dataset is composed of soil parameters for various blocks and villages of Thanjavur district. There are 14 blocks, and each block's soil fertility index namely micro (Zn, B, Fe, Mn and Cu) and macro nutrients (N, P, K, S) constitution is available in the dataset

Table 3. Soil Dataset

S.NO	SOIL ATTRIBUTES
1	BLOCK
2	Lime Status
3	Texture
4	pH
5	Electrical Conductivity (EC)
6	Nitrogen (N)
7	Phosphorous (P)
8	Potassium (K)
9	Sulphur (S)
10	Zinc (Zn)
11	Iron (Fe)
12	Copper (Cu)
13	Manganese (Mn)
14	Boron (B)

Soil parameters include essential macro and micronutrients affecting fertility and pH balance, enabling effective clustering and nutrient-pH modeling for localized recommendations.

B. Stage 2: Predictive Modelling with TP-NP Mapping Logic

Predictive modelling was conducted using the TP-NP Logic Crop Dataset to extract crop details for Stage 3, facilitating greenhouse control. During this stage, a Linear Regression Model was constructed for Temperature vs. PAR, while a Random Forest Regression Model was employed for Temperature, Illuminance, & TSR and PAR. The TP component involved the prediction of PAR based on temperature using the weather dataset model, integrated into the crop dataset via its average temperature feature to compute the average PAR. Similarly, the Nutrients model, developed using Linear Regression and trained on the Soil Dataset, was utilized in the crop dataset's nutrient attributes to estimate the pH value.

$$PAR = 2.97 * Temp + 479.82 \quad (1)$$

From the block wise fertility index in soil dataset, it is identified that only B, Mn, Cu, Fe, Zn and S exhibit a high correlation with pH. Using these variables exclusively, pH is modelled through Linear Regression and derived the following equation to map nutrients to pH.

$$pH = 7.0546 + (-0.4087) * B + (-0.3675) * Mn + (-0.3228) * Cu + (0.2002) * Fe + (1.0895) * Zn + (0.0346) * S \quad (2)$$

The Crop dataset Attributes is composed of 47 different crops grown in Thanjavur district and were selected to reflect both nutrient and climatic requirements, enabling precise crop-soil-environment alignment.

Table 4. Crop Dataset

S.NO	CROP ATTRIBUTES
1	Crop
2	Average Temperature
3	Average Rainfall
4	Rainfall Contribution (%)
5	Relative Humidity
6	Season Classification
7	Nitrogen (N)
8	Phosphorous (P)
9	Potassium (K)
10	Sulphur (S)
11	Zinc (Zn)
12	Boron (B)
13	Iron (Fe)
14	Manganese (Mn)
15	Copper (Cu)

C. Stage 3: Soil clustering and Crop recommendation

In this stage, villages were grouped based on how similar their soil nutrients were, using a method called hierarchical clustering. This helped create soil clusters, where each group had common characteristics. After this, crops that are known to do well in those soil types were linked to the corresponding clusters.

Once this was done, we could take any village, check which soil group it belongs to, and see what crops are most suitable for that group. If a chosen crop fits the recommendation for that cluster, it is considered a good option. However, if the crop does not show up in the recommended list for that soil type, it is marked for review, since it may not grow well there.

This step makes sure that crop choices match local soil conditions, which can really help improve yield and reduce the risk of poor growth.

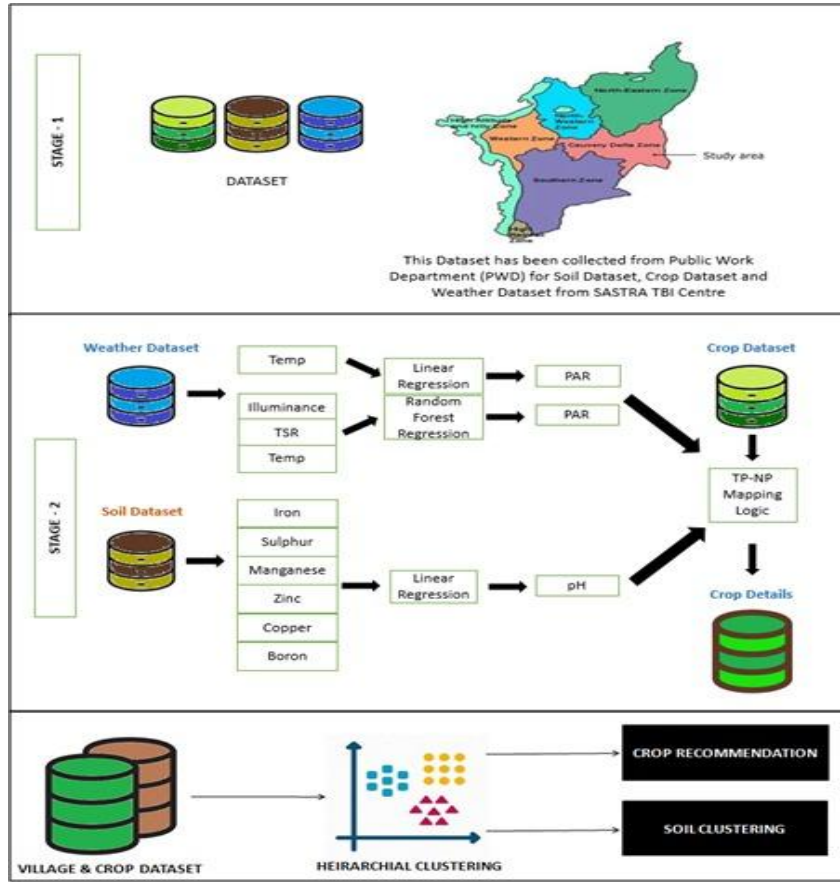


Fig 1. Proposed System Architecture

V. RESULTS AND DISCUSSIONS

A. Relationship between PAR and Key Environmental Variables

In Figure 2, the analysis of PAR and weather parameters are shown.

a. PAR and Temperature

Due to the shared dependence on solar radiation, both PAR and temperature are strongly correlated. Peak solar hours lead to simultaneous increases in temperature and PAR, even though the relationship weakens under cloud cover, where PAR drops more sharply. With higher latitudes and colder seasons showing greater variability, seasonal and geographical factors further modulate this interplay. Therefore, understanding this relationship is critical for optimizing crop productivity.

b. PAR and Tropical Storm Risk (TSR)

TSR is influenced by sea surface temperatures, which drives oceanic heat energy. It also shows a linear relationship with TSR under stable climatic conditions, since PAR is a subset of solar radiation. Monitoring PAR and TSR together provides insights into TSR, which aids understanding this relationship is critical for optimizing crop productivity and photosynthesis.

c. PAR and Illuminance

As both originate from the visible light spectrum (400-700 nm), PAR and Illuminance are closely linked. Illuminance sensors can act as cost-effective proxies for PAR in controlled environments. This correlation is especially useful for greenhouse lighting optimization, even though direct PAR measurements are preferred for precision applications.

d. Model Performance and Forecasting

The PAR analysis integrates linear regression models for parameter predictions. The models R^2 values for predicting PAR using temperature, TSR, and Illuminance is 0.99, which reflects a strong predictive accuracy using Random forest regressor and for predicting PAR using Temperature is 0.79 using linear regression.

B. Relationship between Soil pH and Nutrients

Soil pH plays a very important role in determining the availability and uptake of various nutrients, both macronutrients (N, P, K, S) and micronutrients (Zn, B, Fe, Mn, Cu). This influences chemical solubility and biological processes in the soil.

a. Nutrients, which are highly related to pH

- Iron (Fe): Decreases sharply in alkaline soils due to precipitation and mostly available in acidic soils (pH 4.0–6.5).
- Manganese (Mn): It forms insoluble oxides in alkaline conditions but soluble in acidic soils (pH 5.0–6.5).

b. Nutrients, which are moderately related to pH

- Nitrogen (N): Its availability peaks between pH ranges 6.0–8.0. Alkaline conditions promote nitrification, but efficiency is reduced due to extreme pH levels through volatilization or denitrification.
- Sulfur (S): More mobile and available in acidic soils and less so in alkaline conditions.
- Zinc (Zn): Due to precipitation, solubility decreases in alkaline soils (above pH 7.0), even though it remains moderately influenced compared to other micronutrients.

c. Nutrients, which are weakly related to pH

- Boron (B): Availability decreases slightly in both highly alkaline and highly acidic soils, but it is less influenced by pH than others are.
- Copper (Cu): Even though it is less pH-sensitive than phosphorus or manganese, availability reduces in alkaline soils (above pH 6.5).

C. Greenhouse Crop Cultivation: Optimal PAR and pH Analysis

PAR and pH are crucial environmental factors influencing greenhouse crop productivity. This is derived from analysis evaluates dataset values against few optimal cultivation parameters derived from scientific research for key crops, including tomatoes, cucumbers, green chilies, onions, and lady's fingers with the predictive model. The results highlights alignment and areas for improvement, which provides actionable insights for enhanced greenhouse management practices. Figure 3 presents the comparative analysis of soil pH and nutrients.

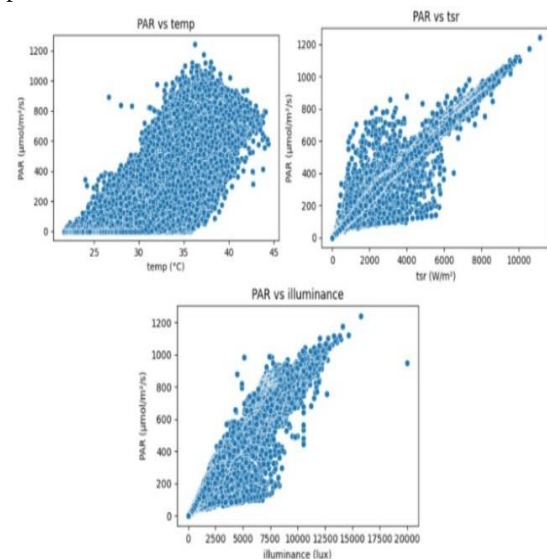


Fig. 2 Comparative Analysis of PAR and Weather Parameters

To address the slight pH deviation for cucumber, the following solutions can be applied to it:

1. Acidic Amendments: Adding sulfur or aluminum sulfate to gradually lower soil pH
2. Organic Matter: Incorporating peat moss or composted manure to naturally acidify the soil.
3. Acidic Fertilizers: promoting soil acidity using ammonium sulfate-based fertilizers.
4. Water Management: Using phosphoric or citric acid, acidify alkaline irrigation water.
5. Mulching: Applying acidic mulches like pine needles or oak leaves.

Regular monitoring and adjustments made will ensure pH aligns with the optimal range.

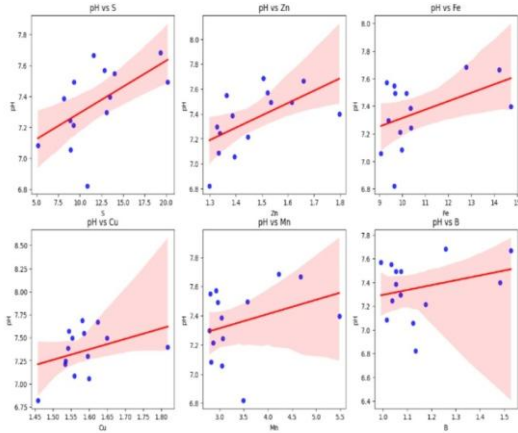


Fig. 3 Comparative Analysis of Soil pH and Nutrients

D. Soil Clustering and Crop Recommendations

Hierarchical clustering was applied to classify villages based on soil nutrient characteristics to boost region-specific agricultural planning. This approach helped group areas with similar soil profiles. Using these clusters, crop recommendations were generated by matching suitable crops to each soil group. This method aids more insightful, localized decision-making and contributes to efficient land use and increased crop yield. Figure 4 and 5 show the village cluster and crop cluster respectively.

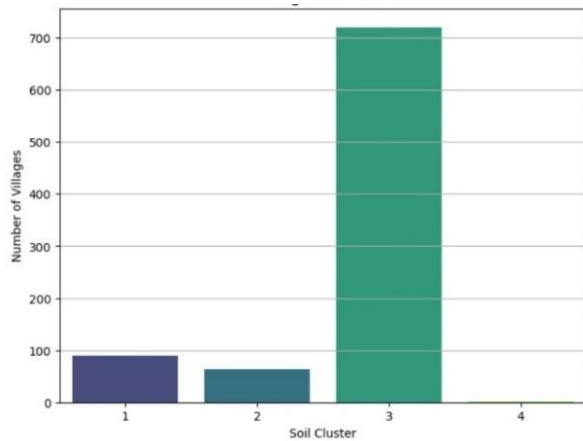


Fig. 4 Number of Villages in Each Cluster

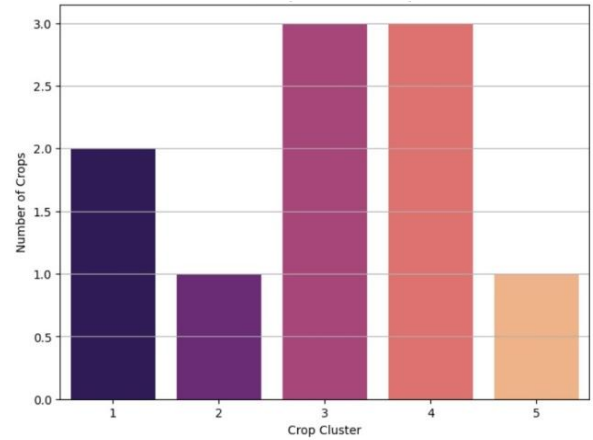


Fig. 5 Number of Crops in Each Cluster

Table 5. Village–Soil Cluster Mapping

Village Name	Soil Cluster	Key Soil Features
Ayyampettai	Cluster 1	High Nitrogen, Moderate pH
Kabisthalam	Cluster 2	Low Phosphorus, Slightly Alkaline
Thirupanandal	Cluster 3	Balanced NPK, Acidic pH
Papanasam	Cluster 1	High Organic Matter
Melattur	Cluster 4	High Potassium, Low Micronutrient Availability

Note: Soil clusters were formed using hierarchical clustering based on nutrient and pH values

VI. CONCLUSION

The above study paper highlights the crucial role that Photosynthetically Active Radiation (PAR) and pH levels can have on maximizing agricultural yield potential within the State of Tamil Nadu's, Cauvery Delta region.

Table 6. Crop Recommendations Based on Soil Clusters

Soil Cluster	Mapped Crop Cluster	Recommended Crops	Ideal pH	Ideal PAR ($\mu\text{mol}/\text{m}^2/\text{s}$)
Cluster 1	Crop Cluster A	Tomato, Brinjal	6.0–6.5	450-650
Cluster 2	Crop Cluster C	Okra, Green Chili	6.2–6.8	500-700
Cluster 3	Crop Cluster B	Onion, Bitter Gourd	5.8–6.4	400-600
Cluster 4	Crop Cluster D	Groundnut, Bottle Gourd	6.5–7.0	480-650

Note: Crop clusters were assigned based on crop nutrient/pH needs and greenhouse PAR ranges

Through the process of employing Machine Learning models like Linear Regression and Random Forest Regression, the research proves strong correlations between determinant environmental factors—temperature, illuminance, and total solar radiation—and soil factors such as pH and nutrient content, all of which contribute significantly to farm productivity.

The PAR prediction was on average a quite accurate R^2 value of 0.99 with temperature, illuminance, and solar radiation as input parameters. Likewise, pH prediction averaged an R^2 value of 0.80 and underscores its critical function in governing the uptake of essential nutrients as well as soil vitality.

The study presents a novel TP–NP mapping (Temperature PAR – Nutrients pH) methodology and proposes an incorporation of cutting-edge AI-based forecasting technologies and monitoring models to maintain and improve environmental and soil conditions sustainably. This methodology allows local PAR and pH values to match global agricultural guidelines, therefore encouraging informed decision making and promoting sustainable agriculture.

In addition to environmental analysis as a means of optimizing crop performance, the research includes the implementation of a Hierarchical Clustering-based crop recommendation system. By clustering villages and soils

based on their environmental and nutrient profiles, the system ensures that the right crops are suggested to each cluster, enhancing adaptability and maximizing production.

In brief, this study proposes a whole-of-farm data-driven approach to achieve sustainable agriculture in Cauvery Delta by integrating predictive modeling, clustering, and environmental analytics to improve agricultural productivity while optimizing the use of limited resources. In the future, this work can be expanded by incorporating real-time satellite data, IoT-based soil monitoring, and mobile-based farmer advisory platforms. This would enable more precise, timely recommendations and make the system scalable to other regions with similar agricultural challenges.

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